Infosys_HotelRevAI – Milestone - 3

TOPIC :

DAX Measures & Business Metrics Documentation for AI-Driven Hotel Revenue Analysis

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Tool Used: Microsoft Power BI

1. Introduction

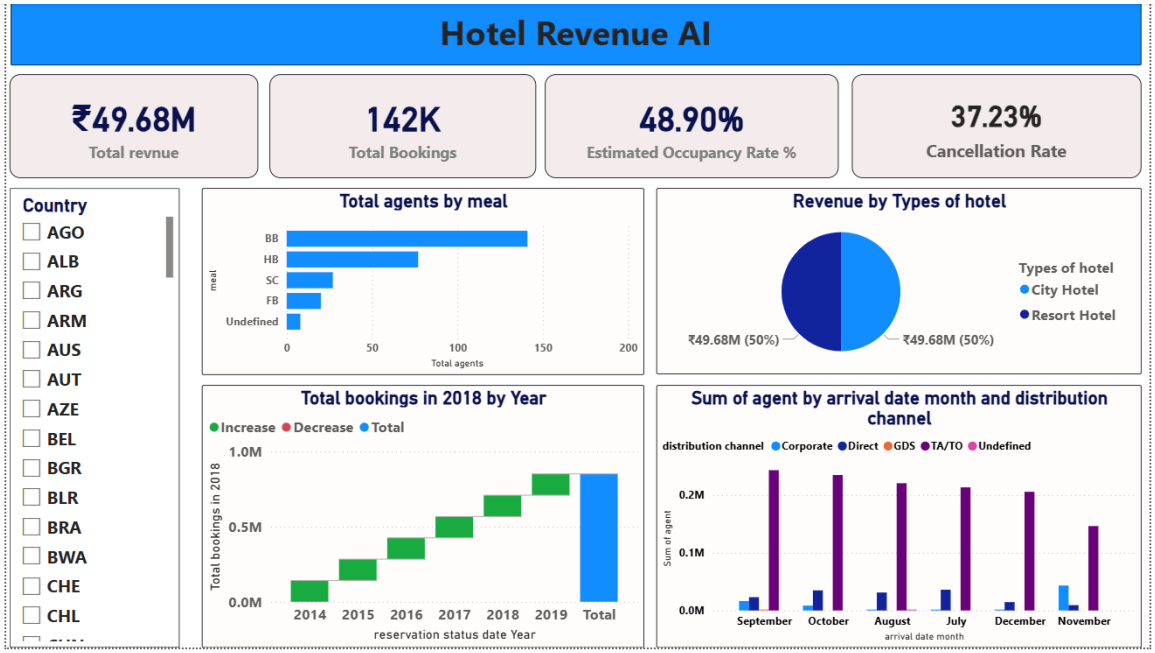
The hospitality industry generates large volumes of booking and customer data every day. However, raw data alone does not help hotel management make effective decisions unless it is properly analyzed and visualized. The Infosys_HotelRevAI project is designed to address this gap by transforming hotel booking data into meaningful insights using Microsoft Power BI.

This project focuses on analyzing hotel revenue, occupancy, customer behavior, and operational performance through interactive dashboards. Advanced analytics techniques such as DAX measures and time-based comparisons are used to provide a clear understanding of business performance.

The main objective of this documentation is to explain the DAX calculations and business metrics used in the project, highlighting their relevance in real-world hotel management scenarios.

2. Project Dashboards Overview

2.1. Hotel Revenue AI (Overview)



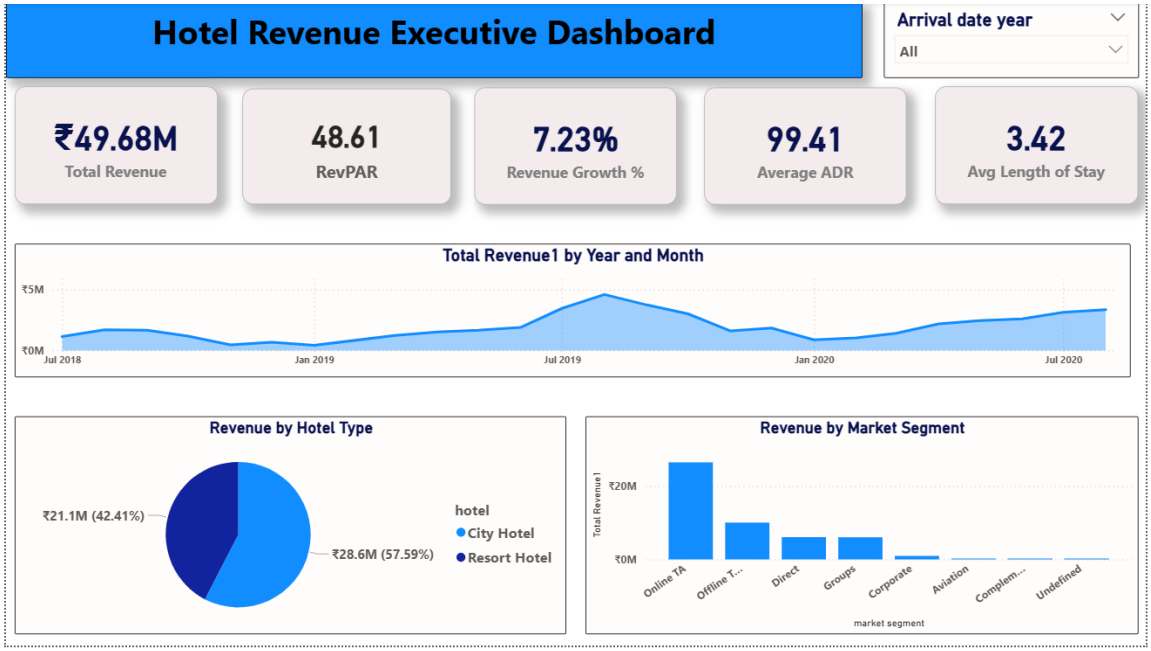
Purpose:
Provides a high-level snapshot of overall hotel performance.

Key Insights:

- Total Revenue
- Total Bookings
- Occupancy Rate
- Cancellation Rate
- Revenue distribution across hotel types

This dashboard is primarily used by management for quick performance monitoring.

2.2. Executive Financial Overview



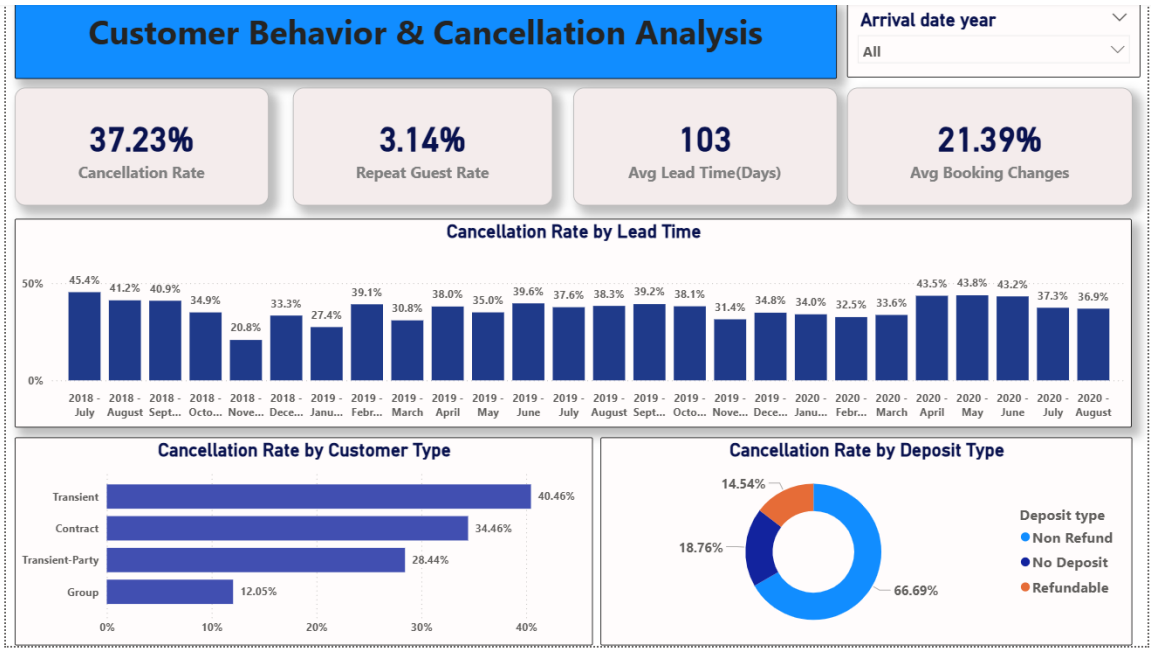
Purpose:
Focuses on strategic revenue and pricing metrics.

Key Insights:

- Revenue Growth %
- RevPAR (Revenue Per Available Room)
- Average Daily Rate (ADR)
- Monthly revenue trends

This dashboard supports executive-level financial planning and performance evaluation.

2.3. Customer & Booking Behavior Analysis



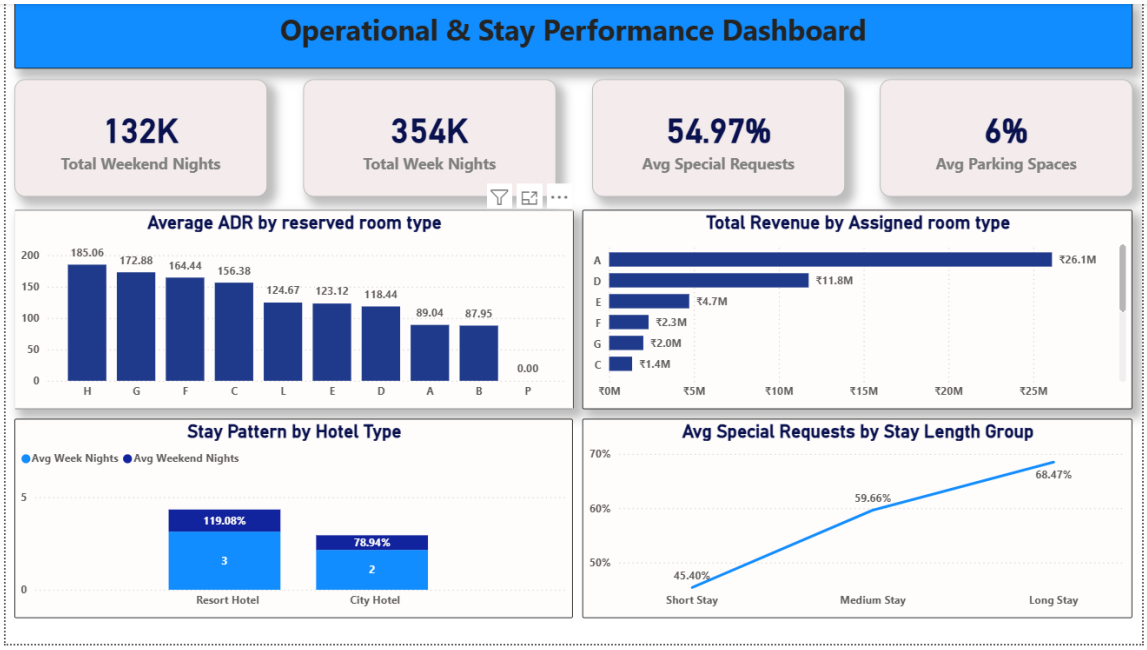
Purpose:
Analyzes customer behavior and booking reliability.

Key Insights:

- Cancellation Rate
- Repeat Guest Rate
- Average Lead Time
- Booking behavior trends

This dashboard helps improve booking policies and customer retention strategies.

2.4. Operational & Stay Performance Dashboard



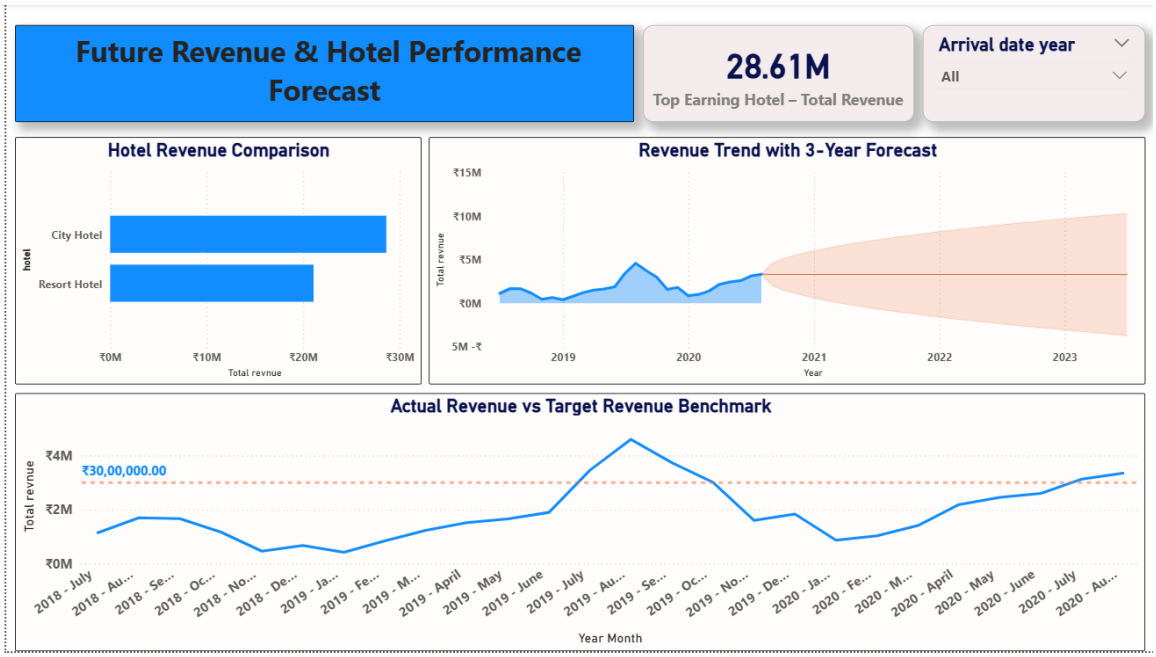
Purpose:
Analyzes operational workload and stay patterns.

Key Insights:

- Weekday vs Weekend stays
- Room utilization
- Special requests
- Parking space usage

This dashboard supports operational planning and resource optimization.

2.5 Future Revenue & Target Performance Dashboard



Purpose:
Focuses on future revenue trends, forecasting, and benchmark comparison.

Key Insights:

- Revenue trend with 3-year forecast
- Actual Revenue vs Target Revenue Benchmark
- Hotel revenue comparison
- Top Earning Hotel KPI
- Expected Revenue (Next 12 Months)

This dashboard supports strategic planning and future decision-making.

3. Core Revenue DAX Measures

DAX measures are the analytical foundation of this project. They allow dynamic calculations that update automatically based on filters and slicers.

3.1 Total Revenue

- Total Revenue is calculated by summing the revenue generated from all bookings. This metric represents the overall financial performance of the hotel.

```
Total Revenue = SUM(Append1[Total_Revenue])
```

3.2 Total Bookings

Business Meaning: Total number of bookings.

```
Total Bookings = COUNT(Append1[reservation_status])
```

3.3 Revenue Growth %

- Revenue Growth Percentage measures the month-over-month change in revenue. It helps management understand whether business performance is improving or declining over time.

```
Revenue Growth % =
VAR CurrentRevenue = [Total Revenue]
VAR PreviousRevenue =
    CALCULATE(
        [Total Revenue],
        DATEADD(Append1[Full Date], -1, MONTH)
    )
RETURN
DIVIDE(CurrentRevenue - PreviousRevenue, PreviousRevenue)
```

4. Operational Performance DAX Measures

4.1 Occupancy Rate %

- Occupancy Rate calculates the proportion of available room nights that are occupied. This metric is critical for evaluating operational efficiency.

```
Occupancy Rate % =
DIVIDE(
    SUM(Append1[stays_in_week_nights]) +
    SUM(Append1[stays_in_weekend_nights]),
```

```
    COUNT(Append1[hotel]) * 7
)
```

4.2 Average Daily Rate (ADR)

- Average Daily Rate (ADR) represents the average price paid per occupied room and is used to analyze pricing strategy.

```
Average ADR = AVERAGE(Append1[adr])
```

4.3 RevPAR

- RevPAR (Revenue Per Available Room) combines ADR and Occupancy Rate to measure revenue efficiency and is considered the most important KPI in hotel analytics.

```
RevPAR = [Average ADR] * [Occupancy Rate %]
```

5. Customer Behavior DAX Measures

5.1 Cancellation Rate %

- Cancellation Rate measures the percentage of bookings that were cancelled, helping identify booking risk and customer reliability.

```
Cancellation Rate % =
DIVIDE(
    [Total Cancelled Bookings],
    [Total Bookings]
)
```

5.2 Average Lead Time

Average number of days between booking and arrival.

- Average Lead Time represents how far in advance customers typically make bookings, which is useful for demand forecasting.

```
Avg Lead Time = AVERAGE(Append1[lead_time])
```

6. Service & Stay Metrics

6.1 Total Week Nights

```
Total Week Nights = SUM(Append1[stays_in_week_nights])
```

6.2 Total Weekend Nights

```
Total Weekend Nights = SUM(Append1[stays_in_weekend_nights])
```

6.3 Total Special Requests

```
Total Special Requests = SUM(Append1[total_of_special_requests])
```

7. BUSINESS METRICS & KPI INTERPRETATION

This section focuses on the business significance of the metrics rather than their calculations. These KPIs help hotel management make informed strategic and operational decisions.

7.1 Strategic Revenue Metrics

- **Total Revenue** represents the overall financial performance of the hotel.
- **Revenue Growth %** indicates whether the hotel's revenue is improving or declining over time.
- **RevPAR** measures how effectively available rooms are generating revenue and is considered a key indicator of hotel profitability.

These metrics are mainly used by **executive management** to assess business health.

7.2 Operational Performance Metrics

- **Occupancy Rate** shows how efficiently hotel rooms are being utilized.
- **Total Week Nights and Weekend Nights** help distinguish between business travel demand and leisure travel demand.
- Higher weekday stays usually indicate corporate bookings, while higher weekend stays reflect leisure tourism.

These metrics support decisions related to **staffing, pricing, and capacity planning**.

7.3 Customer Behavior Metrics

- **Cancellation Rate** helps identify booking risk and reliability.
- **Average Lead Time** shows how early customers plan their bookings.

- **Repeat Guest Rate** reflects customer loyalty and satisfaction.

Understanding these metrics allows hotels to improve **booking policies and marketing strategies**.

7.4 Service Usage Metrics

- **Special Requests** indicate guest service expectations.
- **Parking Spaces Usage** reflects facility demand and infrastructure utilization.

Higher values suggest the need for enhanced service management and operational readiness.

8. Future Revenue & AI-Driven Forecasting KPIs

8.1 Top Earning Hotel – Total Revenue

```
Top Earning Hotel - Total Revenue =

MAXX(

    VALUES(Append1[hotel]),

    [Total Revenue]

)
```

Business Value:
Identifies the strongest revenue contributor.

8.2 Expected Revenue (Next 12 Months)

```
Expected Revenue (Next 12 Months) =
CALCULATE(
    [Total Revenue],
    DATESINPERIOD(
        Append1[Full Date],
        MAX(Append1[Full Date]),
        12,
        MONTH
```

```
    )
)
```

Business Value:
Provides short-term revenue outlook, critical for executive planning.

9. Conclusion

The Infosys_HotelRevAI project demonstrates the effective use of Microsoft Power BI, DAX, and AI-assisted analytics to analyze hotel revenue, operations, and customer behavior. By integrating business KPIs with interactive dashboards, the project enables data-driven decision-making across strategic, operational, and customer-focused areas.

The inclusion of forecasting and benchmark comparison enhances the project's analytical depth and aligns it with real-world business intelligence solutions used in the hospitality industry.